Richard Gonzalez

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SUMMARY

Full-stack software engineer who builds software with the people it serves. Spent 3+ years at General Motors meeting twice a week with vehicle program analysts and turning their pain points into features, like a 20-second wait cut to 2. In 2026, independently delivered a complete AI-assisted web app, from the user interface to the AWS hosting.

TECHNICAL SKILLS

Languages: Java, TypeScript, JavaScript, SQL
Frontend: Angular, RxJS, PrimeNG, Angular Material, AG Grid, Tailwind CSS
Backend & Data: Spring Boot, Spring Security, Spring AI, Spring Data JPA, REST APIs, PostgreSQL, pgvector, Redis, Flyway
Cloud & DevOps: AWS (EC2, S3, CloudFront, SQS, Cognito, ECR, IAM), Terraform, Docker, Kubernetes, Git, GitHub Actions (CI/CD)
Testing & Observability: JUnit 5, Testcontainers, Jest, Playwright, JMeter, CloudWatch, OpenTelemetry, Grafana

EXPERIENCE

Software Engineer Jan 2023 – May 2026
General Motors *Austin, TX*

One of 5 developers on internal tools ~300 GM vehicle program analysts use to plan the features and options of 40+ programs.

- Slashed vehicle catalog load times 90% (~20 s to 2 s; trucks ~2 min to 20 s) by tuning one slow database query, sparing analysts an estimated 1,000+ hours of waiting per year (PostgreSQL, JMeter).
- Halved analysts' manual data entry with a feature that mirrors each paired vehicle-option rule, keeping mismatched rules out of about a dozen downstream GM systems (Angular, Spring Boot, PostgreSQL).
- Migrated a vehicle-planning app onto GM's in-house hosting platform ahead of a deadline carrying $1M+ in penalties; three other teams then adopted the deployment setup and release pipeline created for it (Kubernetes, CI/CD).
- Closed a must-fix security flag on the planning tool ~300 analysts rely on, single-handedly modernizing its outdated core frameworks and repairing everything the upgrade broke (Angular 15 to 18, Spring Boot 2.7 to 3.4).
- Hardened every release with automated tests that block a build when a core workflow fails: the frontend's first tests (0% to 50% coverage) and backend coverage from ~20% to 90% across seven services (Jest, Playwright, JUnit).

PROJECTS

Vehicle Catalog | catalog.rgonz.dev | github.com/rgonzal92/vehicle-catalog 2026
Java, Spring Boot, Spring AI, Angular, PostgreSQL, Redis, AWS, Terraform

A solo, from-scratch vehicle catalog app that defines each vehicle's features, live as a public demo: edit a catalog like a spreadsheet, submit changes for approval, track every version, and have AI propose rules. 24 design records, 1,100+ tests on GitHub.

- Designed a spreadsheet-style editor that saves each change as you type, even at 48,000 cells, with a manager's approval required before any version becomes official; stress-tested it at full size and removed a half-second delay from every save.
- Shipped AI that suggests catalog rules and answers questions about catalogs and uploaded files, passing 81 of 81 tests; every rule it writes is checked by the app and signed off by a person, within a $1/day budget (OpenAI, RAG, pgvector).
- Scripted hands-off releases on AWS: each update goes live by itself and reverts to the last working version if it doesn't start correctly; proved the database can be restored on a new server in 7 minutes (GitHub Actions, OIDC).
- Offloaded slow tasks (exports, AI summaries, document processing) to the background so no page stalls on them; crashed tasks are retried, then set aside for review, not lost, and the app stays up when a supporting service drops out (SQS).
- Eliminated error pages during releases by having the server pause incoming requests until the new version is ready, confirmed by a once-a-second test with zero errors; then added dashboards and email alerts (CloudWatch).

EDUCATION

The University of Texas at San Antonio San Antonio, TX
Bachelor of Science in Computer Science *May 2022*